Global CO_2 Emissions: Trends, Regional Shifts, and Per Capita Ine

1750–2024 | Data-forward briefing for policy analysts and climate negotiators

Scope: long-run emissions growth, shifting regional responsibility, a nequali

Anchor sources: Global Carbon Budget 2025 and NASA atmospheric CO_2 records.

429 ppm

(Feb 2026)

Sources: Global Carbon Budget 2025; NASA

Atmospheric CO_2 Reaches 429 ppm — 50% Above Pre-Industrial

429 ppm

Mauna Loa measurement
February 2026

Pre-industrial baseline: ~280 ppm.

Current level is ~50% higher than pre-industrial.

Instrumental record at Mauna Loa begins in 1958.

Long-run acceleration aligns with fossil-fuel era emissions growth.

Source: NASA / Scripps Mauna Loa record; contextual baseline from source brief

Global Fossil Fuel CO_2 Emissions Grew Six-Fold: 6 Billion to 35+ B

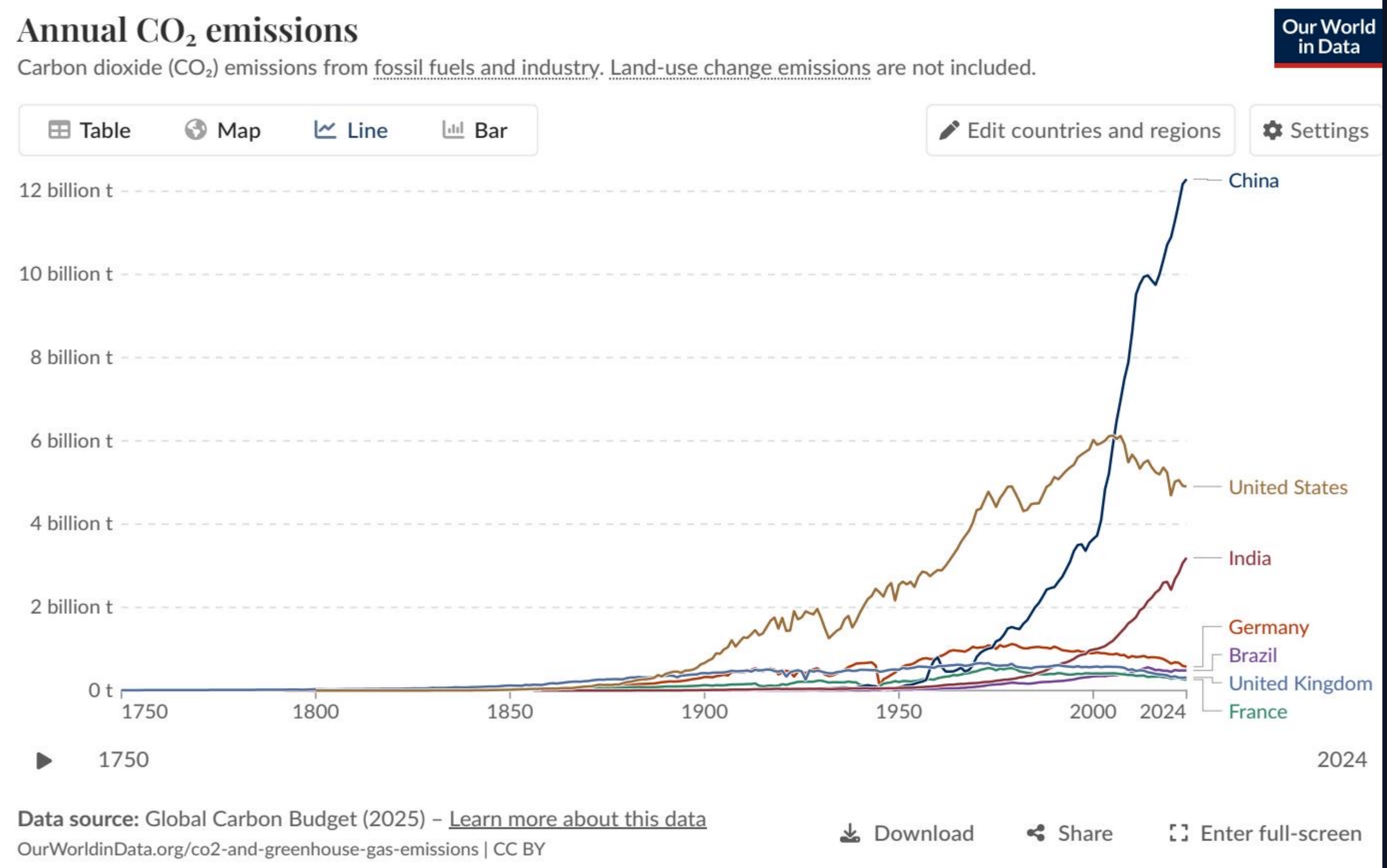
Key inflection points:

- ~6 Bt in 1950
- ~20 Bt by 1990
- >35 Bt by 2024

Trend: growth slowed in rate, but total emissions

Reference visual: figure_2.jpg (source brief)

China Now Emits Over 12 Billion Tonnes — More Than Double th

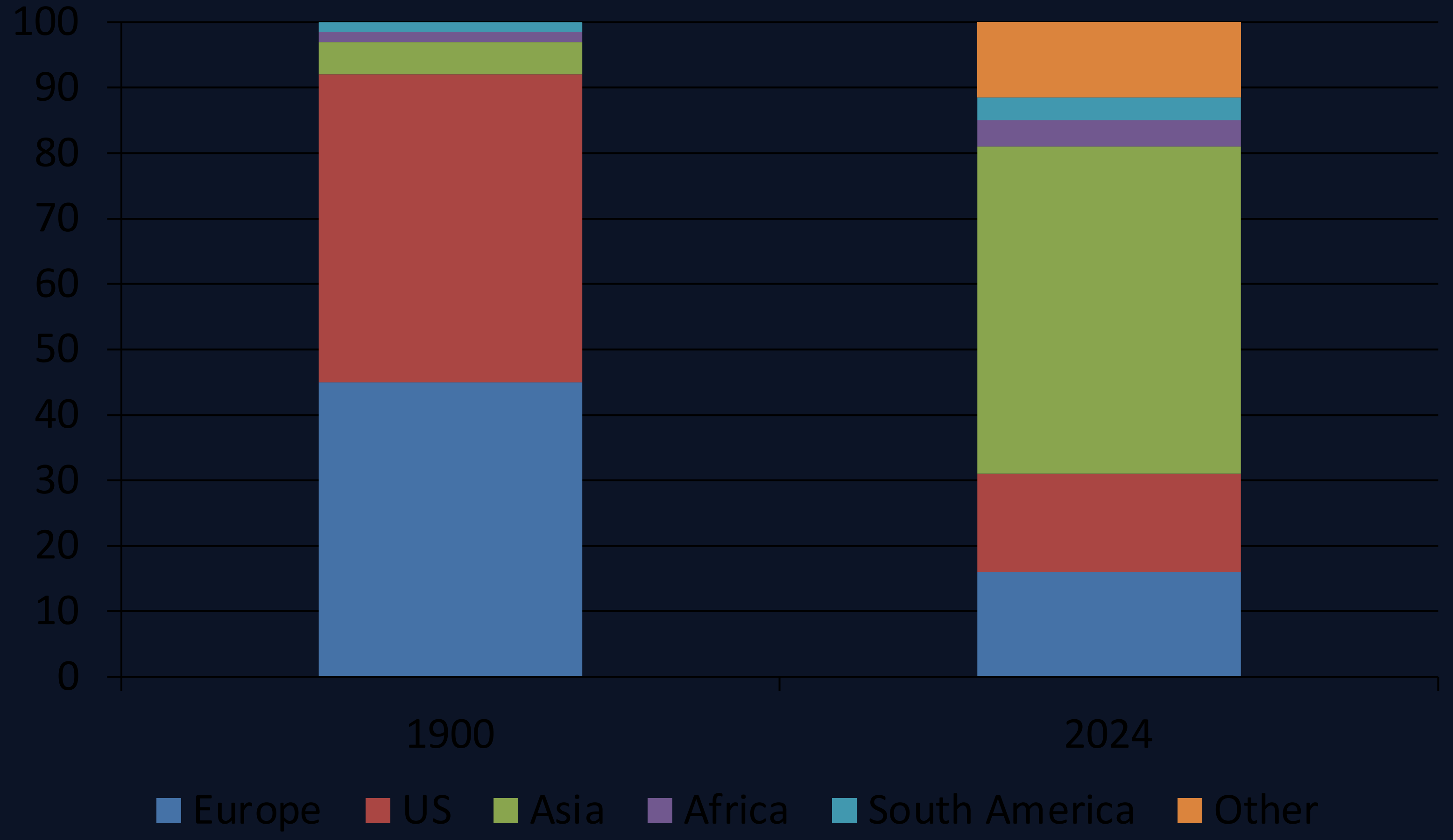


Latest annual totals (approx.):

China: ~12.5 Bt
United States: ~5 Bt
India: ~3 Bt
Germany: ~0.6 Bt | UK: ~0.35 Bt
Brazil: ~0.5 Bt | France: ~0.3 Bt
China's rise post-2000 reshaped global totals; US below

Reference visual: figure_3.jpg (source brief)

Asia Now Accounts for ~50% of Global Emissions; US + Europe Be



Shift in regional shares:

1900: Europe + US >90%

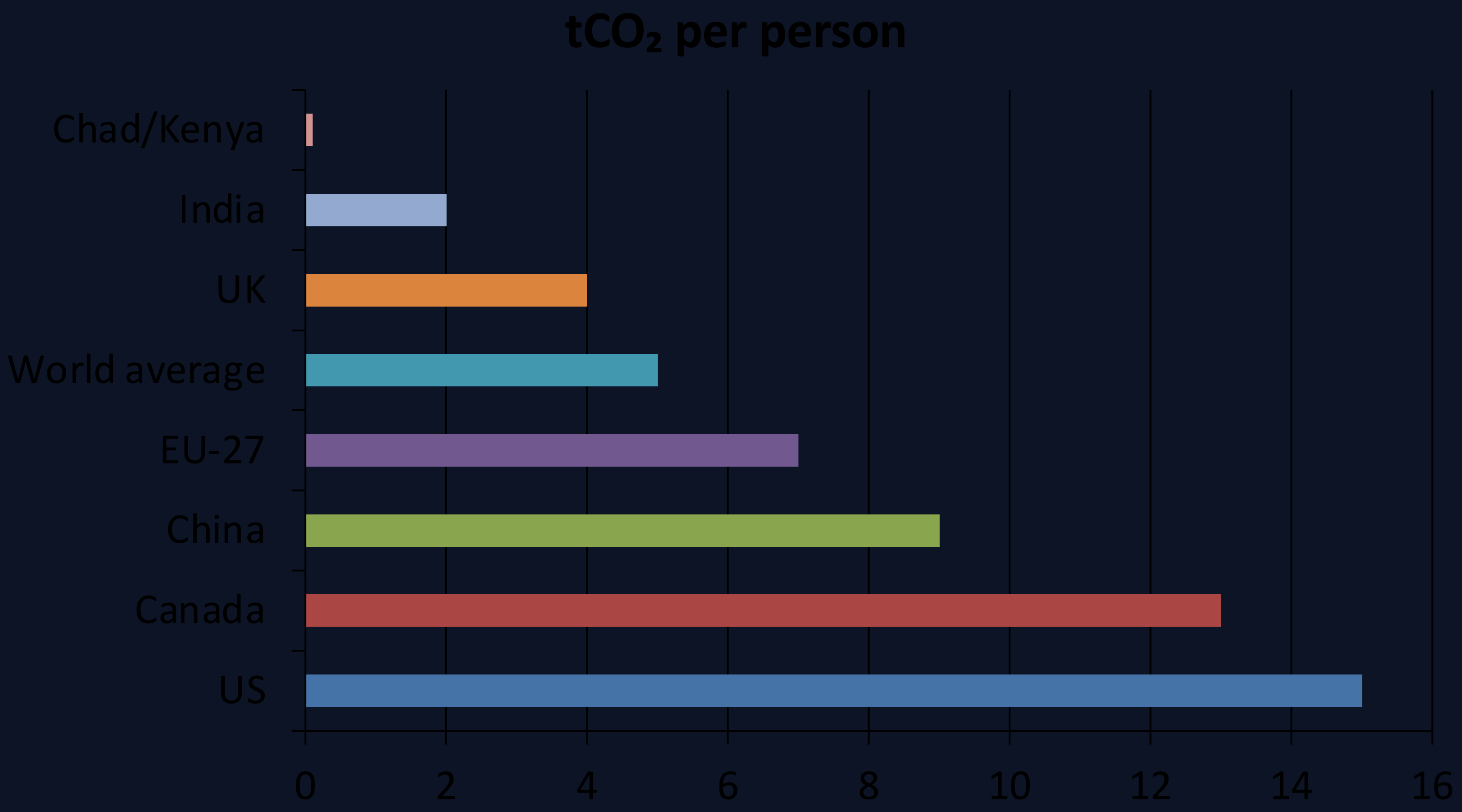
2024: Asia ~50%

US + Europe: <33% combined

Africa + South America: ~3–4% each

Sources: Global Carbon Budget 2025; NASA

Per Capita Gap: 15 t in the US vs. 0.1 t in Chad — a 150× Disparit

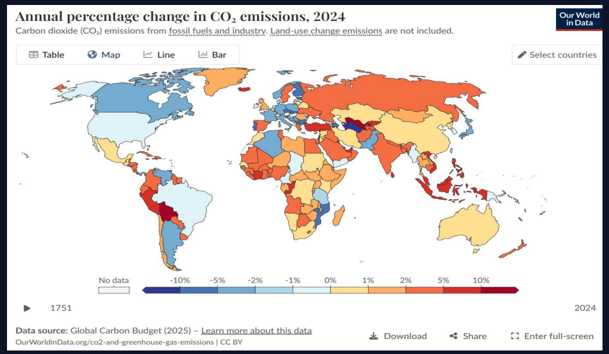


Disparity spotlight

Highest vs lowest in this comparison: 15 ÷ 0

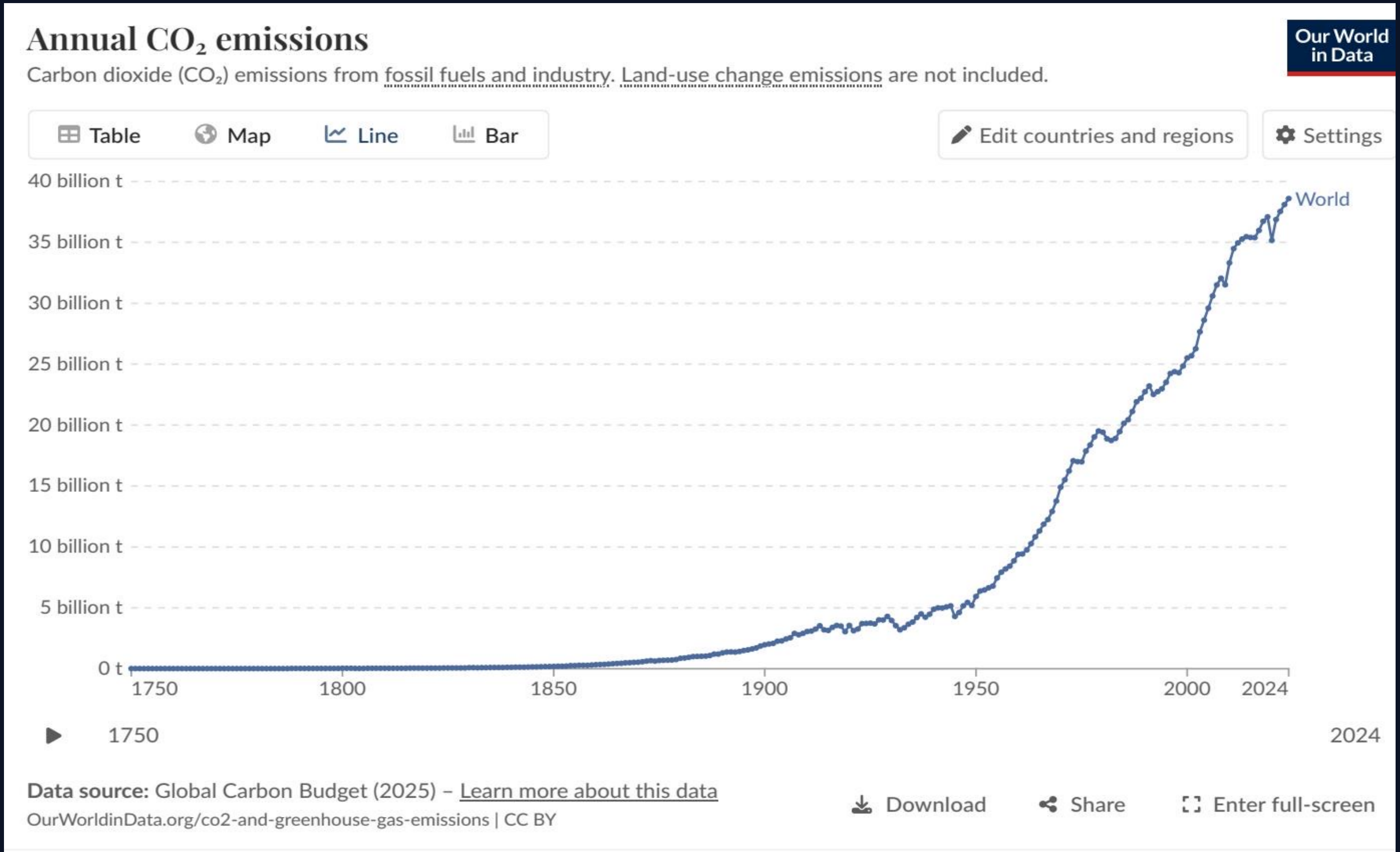
High per-capita values remain concentrated in hig

Reference trend visual: figure_1.jpg



Reference visual: figure_1.jpg (source brief)

2024 Snapshot: US and Europe Cut Emissions While Russia and S



Map interpretation (2024):

US + much of Europe: blue (−1% to −5%)

Russia + parts of SE Asia: orange/red (+2% to +10%+)

Several African regions also show increases.

Pattern indicates diverging transition speeds across re

Reference visual: figure_4.jpg (source brief)

Energy Mix Explains Why France Emits Far Less Per Capita Than G

Countries with similar living standards can diverge strongly on per-capita CO_2 depending on electricity mix, heating fuels, and industrial structure.

Country pair	Dominant low-carbon sources	Higher-carbon reliance	Per capita CO_2 (approx.)
France	Nuclear + renewables	Lower fossil share	~4–5 t/person
UK	Gas + growing renewables	Lower coal than peers	~4 t/person
Germany	Renewables + coal/gas legacy	Higher fossil share than France	~7–8 t/person
Netherlands	Gas-intensive system	Fossil-heavy industry/heat	~8–9 t/person

Sources: Global Carbon Budget 2025; NASA

Key Takeaways: Emissions Are Still Rising, But the Map of Respo

Global emissions exceed 35 Bt and have not yet peaked.

China contributes >25% of annual global CO_2, with India still rising.

Per-capita inequality remains extreme: roughly 150× from highest to lowest emitters.

US and Europe are declining in recent annual flows, but cumulative historical responsibility remains large.

Policy and energy mix choices—not prosperity alone—drive large differences in per-capita outcomes.

Sources: Global Carbon Budget 2025; NASA